

Bentley Oakes

Assistant Professor

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Key Academic Skills

Researcher: Lead- and co-author of dozens of publications with hundreds of citations on Google Scholar.

Collaborator: Involved in multiple international academic and industrial collaborations.

Teacher/Mentor: Extensive experience teaching courses and mentoring graduate students.

Community-Builder: Organizing and program committee member for local and global events.

Language Skills

English: Native proficiency

French: Intermediate

Education

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|------------|--|------------------------------|
| PhD | McGill University , Computer Science | Montréal, Canada
Jan 2019 |
| | <ul style="list-style-type: none"> Thesis: <i>A Symbolic Execution-Based Approach To Model Transformation Verification using Structural Contracts</i> Supervisors: Hans Vangheluwe and Clark Verbrugge | |
| MSc | McGill University , Computer Science | Montréal, Canada
Jan 2013 |
| | <ul style="list-style-type: none"> Thesis: <i>Practical and Theoretical Issues of Evolving Behaviour Trees for a Turn-Based Game</i> Supervisor: Clark Verbrugge | |
| BSc | University of Manitoba , Computer Science (Honours) | Winnipeg, Canada
Jan 2011 |
| | <ul style="list-style-type: none"> Co-op internship at Electronic Arts Inc., Montréal — Fall 2009, Summer 2010: Prototyping artificial intelligence in commercial video games. Co-op internship at Blackberry Limited (RIM), Waterloo — Winter 2009: Implementing cryptographic communication protocols. | |

Research Experience

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| Polytechnique Montréal , Assistant Professor | Montréal, Canada
Sept 2023 – present |
| <ul style="list-style-type: none"> Department of Computer and Software Engineering (GIGL) Research topics: Accelerating the construction and deployment of digital twins through ontologies for capturing domain knowledge, and large language models and agentic AI for automating their creation. Integrating formal methods to verify digital twin correctness, and assisting users with the configuration and observability of their machine learning and verification/validation services. | |
| Université de Montréal , Post-Doctoral Researcher | Montréal, Canada
Sept 2021 – Aug 2023 |
| <ul style="list-style-type: none"> Lab: GEODES Software Engineering Research Group Supervisors: Houari Sahraoui and Michalis Famelis Research topic: Assisting non-machine learning experts in constructing ML solutions by exploring tailoring of computational workflows. Research topic: Extracting and structuring developer rationale from open-source commit messages using NLP and Large Language Models, in collaboration with PhD student Mouna Dhaouadi. | |

University of Antwerp, Post-Doctoral Researcher

- Labs: Antwerp Systems and Software Modelling, and Constrained Systems Lab
- Supervisors: Hans Vangheluwe and Joachim Denil
- Research topics: Verification and validation of cyber-physical systems, model-driven engineering, multi-paradigm modelling, co-simulation, and digital twins.

Antwerp, Belgium
Sept 2018 – July 2021

Université de Montréal, Visiting Researcher

- Host: Eugene Syriani, GEODES Software Engineering Research Group
- Research topic: Developing an interface between the AToMPM modelling tool and the ModelVerse modelling repository.

Montréal, Canada
May 2018

fortiss GmbH, Visiting Researcher

- Host: Levi Lúcio
- Research topic: Formalizing representations of model transformation languages.

Munich, Germany
July 2016 – Aug 2016

General Motors Technical Center, Visiting Researcher

- Host: Ramesh Sethu
- Research topics: Applying model transformations for code/model modernization at an industrial scale, and industrial intellectual property concerns.

Warren, USA
Oct 2014 – Dec 2014

Industrial Collaborations

Framework for Systematic Design of Digital Twins (DTDesign)

2019–2021

- Collaborators: Atlas Copco, Flanders Make
- Topics: Digital twin characteristics, integrating knowledge graphs and digital twins.
- Funding: Flanders Innovation and Entrepreneurship Agency (VLAIO)

Automated & Simulation-based Functional Safety Engineering Methodology (aSET)

2018–2020

- Collaborators: Dana Belgium NV, Siemens Industry Software (Leuven), Flanders Make
- Topics: Verification of safety-critical cyber-physical systems (formal methods, reinforcement learning-based fault injection for hazard discovery, simulation/visualization), DevOps for functional safety.
- Funding: Flanders Innovation and Entrepreneurship Agency (VLAIO)

Innovation in the Development of Electrical Systems For Aeronautics (INES)

2019–2020

- Collaborators: Boeing Research & Technology Europe (Madrid), Siemens Industry Software (Leuven), Flanders Make
- Topics: Co-simulation configuration, fault injection in co-simulation.
- Funding: Flanders Innovation and Entrepreneurship Agency (VLAIO)

Network for the Engineering of Complex Software-Intensive Systems for Automotive Systems (NECSIS)

2013–2016

- Collaborators: General Motors of Canada Ltd., Queen's University, University of Antwerp
- Topic: Verification of model transformations.
- Funding: \$14M from Automotive Partnership Canada and Natural Sciences and Engineering Research Council of Canada (NSERC)

Teaching Experience

Polytechnique Montréal, Course Instructor

Montréal, Canada
Jan 2024 – present

- Engineering of Digital Twins (LOG8421E) — Winter 2025, Winter 2026
- Introduction à la programmation (INF1007) — Fall 2025, Fall 2026
- LOG8371E
- Software Quality Engineering — Fall 2024
- INF6900AE/INF7900A
- Scientific and Technical Communication — Winter 2024, Fall 2024, Fall 2025, Fall 2026

Polytechnique Montréal , Guest Lecturer • LOG8704 • XR Software Development — Nov. 2024: <i>Digital Twins: Introductions and Intersections</i> • LOG6953DE • Model-Driven Software Engineering — Nov. 2022, Nov. 2023: Model-driven engineering, usage and verification of model transformations	Montréal, Canada Jan 2022 – Jan 2024
McGill University , Course Lecturer/Coordinator • COMP 202 • Foundations of Programming — Winter 2015, 2017, 2018 (avg. 189 students) • Developed and presented lecture material on Java programming for students with no prior programming experience. • Created assessments and marking guides; coordinated with other instructors and TAs.	Montréal, Canada Jan 2015 – Jan 2018
University of Antwerp , Teaching Assistant • 2001WETMTR • Model-Driven Engineering — Fall 2020 • 2001WETMSI • Modelling of Software-Intensive Systems — Fall 2019	Antwerp, Belgium Jan 2019 – Jan 2020
McGill University , Teaching Assistant • COMP 202 • Foundations of Programming (×2) • COMP 250 • Introduction to Computer Science (×2) • COMP 251 • Data Structures and Algorithms (×3)	Montréal, Canada Jan 2012 – Jan 2014

Student Supervision

Kérian Fiter — PhD Student • Digital twin federation, trust, and observability • Ontological federation and reasoning, large language models, and time-series foundation models	2024–present
Carlos Pambo — PhD Student • Recommendations and patterns for digital twin engineering • Ontologies for digital twin engineering	2024–present
Adil Lagrou — PhD Student • Formalization of digital twin component interoperability • Ontological federation and reasoning for interoperability	2026–present
Gabrielle Gallant — Master’s Student • AI-enhanced digital twins for vibration analysis in manufacturing • Machine learning techniques including time-series foundation models • Co-supervised with Oguzhan Tuysuz	2025–present
Angelica Portocarrero — Master’s Student • Structured digital twin engineering with LLMs • Previously Mitacs Globalink Research Intern (2025)	2026–present
Fatima Aguel — Mitacs Globalink Research Intern • Ontological modelling of digital twin component repositories	Summer 2026

Mikkel Schmidt Andersen — Research Intern • Probabilistic digital twin update scheduling	Jan 2025
Louis Malassigné-Onfroy — Research Assistant • Visualization of digital twin architectures	Jan 2024
Amélie Bosphore — Research Assistant • Implementation of modelled machine learning pipelines	Jan 2024
Justin Caisse — Research Assistant • Ontological modelling of machine learning pipelines	2023–2024

Invited Talks and Presentations

Guest lecture — Engineering Digital Twins course, McMaster University • Host: Prof. Istvan David • <i>What Does Your Digital Twin Do? A Framework and Tooling for Systematic DT Reporting</i>	Mar. 2026
Invited talk — SIG LLODIA (Lecture #3) • <i>Towards Ontology-based Digital Twin Service Construction and Reporting</i>	Jan. 2026
Guest lecture — RAISE course, UCLA • Host: Dr. Maged Elaasar • <i>What is a ‘Digital Twin’ and How Do I Build One?</i>	Nov. 2025
Tutorial — Software Engineering for Machine Learning Applications (SEMLA) • <i>What is a ‘Digital Twin’ and How Do I Build One?</i>	Jun. 2025
Invited talk — EDT.Community • <i>Systematic Digital Twin Reporting</i> • Video recording	May 2025
Invited talk — Next-Generation Cities Institute, Concordia University • <i>Accelerating Digital Twin Research and Engineering</i>	May 2025
Conference paper presentation — ICSE 2025, Ottawa • <i>Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice</i> (journal-first)	May 2025
Guest lecture — Engineering Digital Twins course, McMaster University • Host: Prof. Istvan David • <i>Accelerating Digital Twin Research and Engineering</i>	Mar. 2025
Research presentation — KARI Delegation, Polytechnique Montréal • Advanced Air Mobility (AAM) challenges and advancements	Feb. 2025
Invited talk — Software Engineering at Montréal (SEMTL) • <i>Recent Work on Systematic Reporting and Construction for Digital Twins</i>	Jan 2025
Guest lecture — LOG8704 XR Software Development, Polytechnique Montréal • <i>Digital Twins: Introductions and Intersections</i>	Nov. 2024
Conference paper presentation — MODELS 2024 • <i>Fault localization in DSLTrans model transformations by combining symbolic execution and spectrum-based analysis</i> (SoSyM journal-first)	Sep. 2024

Invited talk — onto:Nexus Forum 2024 • <i>Accelerating Digital Twin Construction with Ontologies</i> • 1st International Forum on Ontological Modeling and Analysis, NASA JPL (attended online)	Jan. 2024
Invited talk — Polytechnique Montréal • <i>Bridging the Gap in Verification and Validation of Complex Systems</i>	Jan 2023
Guest lecture — LOG6953DE, Polytechnique Montréal • Host: Prof. Mohammad Hamdaqa • <i>Model/Graph Transformations: Specification and Verification</i>	Nov. 2022
Invited talk — Software Engineering at Montréal (SEMTL) September seminar • <i>Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice</i>	Sep. 2022
Invited talk — Concordia University • Host: Prof. Essam Mansour • <i>Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice</i>	Jan 2022
Invited talk — Université du Québec à Montréal • <i>Capturing and Utilising Domain-Specific Knowledge</i>	Jan 2022

Community Building

Workshop Organizer — onto:Nexus Workshop 2026 (International Workshop on Ontological Modeling and Analysis) • Co-located with MODELS 2026, Malaga, Spain. • Co-organizers: Maged Elaasar (Chair), Wahab Hamou-Lhadj, Mohammad Hamdaqa, Edward Kamburjan	Jan 2026
Workshop Organizer — onto:Nexus Workshop 2025 (International Workshop on Ontological Modeling and Analysis) • Co-located with MODELS 2025, Grand Rapids, MI. Hybrid event with IEEE proceedings. • Co-organizers: Maged Elaasar (Chair), Wahab Hamou-Lhadj, Mohammad Hamdaqa	Oct. 2025
Lead Organizer — Software Engineering at Montréal (SEMTL) • Regular seminars for the software engineering researchers in Montréal (~40 attendees per meeting, ~30% are professors). • Coordinating with meeting hosts on content, venue, date, and maintaining website.	Montréal, Canada Aug 2022 – present
Organizing Committee Member • SEMLA 2026 — General Chair • FSE 2026 — Student Volunteer Co-Chair • MODELS 2025 — Student Volunteer Co-Chair • ICSE 2025 — Posters Co-Chair • SEMLA 2024, 2025 — Posters Co-Chair • EDTconf 2024 — Publicity Co-Chair • ANNSIM 2024 — Proceedings Co-Chair • ANNSIM 2022, 2023 — Cyber-Physical Systems Track Co-Chair • MODELS 2022 — Posters Co-Chair	2022–2026

Program Committee Member • ASE Research Track — 2026 • WSC — Simulation in Digital Twins Track — 2026 • ASE NIER Track — 2025 • MODELS Artifact Evaluation Track — 2026 • MODELS — 2025 • ANNSIM — 2021–2024, 2026 • DIGITA Workshop (PerCom) — 2026 • IDETWIN Workshop (CCNC) — 2026 • International Workshop on Architecting and Engineering Digital Twins (AEDT) — 2025, 2026 • EMMSAD — 2025 • EDTconf — 2024, 2025, 2026 • MDE Intelligence — 2022, 2023, 2024 • International Workshop on Models and Evolution — 2022, 2026 • ACM Student Research Competition — 2022 • Spring/Summer Simulation Conference — 2019, 2020	2019–2026
Journal Reviewer • KI – Künstliche Intelligenz — 2026 • Journal of Systems & Software (JSS) — 2025 • Information and Software Technology (IST) — 2025 • Journal of Software and Systems Modeling (SoSyM) — 2020, 2021, 2023, 2025, 2026 (Top 1% Reviewer 2020, 2021; Best Reviewer Award 2025) • ACM Transactions on Software Engineering and Methodology (TOSEM) — 2024 • SIMULATION — 2023, 2024 (Best Reviewer Award June 2023) • Science of Computer Programming (SCP) — 2023 • Journal of Computer Languages (JCL) — 2022 • Journal of Object Technology (JOT) — 2022, 2026 • Empirical Software Engineering (EMSE) — 2022 • IEEE Transactions on Automation Science and Engineering (T-ASE) — 2021	2020–2026
Guest Editor for Journal Special Issue • SIMULATION — <i>Modeling and Simulation for Software-Intensive Systems: from IoT to Digital Twins</i>	Jan 2023
Jury Member • INSA Lyon / Polytechnique Montréal — PhD Thesis Jury Member (2026) • Polytechnique Montréal — Examen de synthèse (doctoral qualifying exam) President (2026) • Polytechnique Montréal — M.Sc. Thesis Jury President (2026) • Polytechnique Montréal — PhD Thesis Jury President (2026) • Polytechnique Montréal — M.Sc. Thesis Jury President (2024) • McGill University — PhD Thesis External Reviewer (2024) • McGill University — M.Sc. Thesis External Reviewer (2023)	2023–2026
Grant Reviewer • NSERC Idea to Innovation grant — 2024 • NSERC Discovery grant — 2025, 2026 • Dutch Research Council (NWO) — 2026	2024–2026
Panelist • ANNSIM PhD Colloquium Panel — 2023	Jan 2023

Departmental Service

2024–Present

- Committee Member, Committee Rayonnement (Communications committee), Department of Computer and Software Engineering, Polytechnique Montréal

Scholarships and Awards

SoSyM Best Reviewer Award • Journal of Software & Systems Modeling	Jan 2025
Simulation Journal Best Reviewer Award	June 2023
SoSyM Top 1% Reviewer • Journal of Software & Systems Modeling	2020, 2021
Best Student Paper Award • SIMULTECH — for the paper <i>HintCO – Hint-based configuration of co-simulations</i>	Jan 2019
NSERC Postgraduate Scholarship – Doctoral (PGS D) • Natural Sciences and Engineering Research Council of Canada	2015–2016
Lorne Trottier Science Accelerator Fellowship • McGill University	2015, 2016
Harold H. Helm Fellowship • McGill University	2013, 2014
Grad Excellence Award in Computer Science • McGill University	2012, 2014

Selected Publications

Bridging the Semantic Web and Model-Based Systems Engineering with the Ontological Modeling Language Maged Elaasar, Bentley Oakes , Eduard Kamburjan, Mohammad Hamdaqa, Abdelwahab Hamou-Lhadj (International Semantic Web Conference (ISWC) — accepted)	Oct 2026
Involving before Evolving: A Vision for Trustworthy Enterprise Digital Twin Engineering Kérian Fiter, Adil Lagrou, Bentley Oakes , Franck Dervault 10.1145/3837062.3839094 (MODELS Companion (EDTconf))	Oct 2026
Model-based digital twin engineering: insights, challenges, and future directions Philipp Zech, Souvik Barat, Benjamin Nast, Bentley Oakes , Judith Michael, Steffen Zschaler, Balbir Barn, Ruth Breu 10.1007/s10270-026-01368-8 (Software and Systems Modeling (SoSyM))	May 2026
Automated Extraction and Analysis of Developer’s Rationale in Open Source Software Mouna Dhaouadi, Bentley Oakes , Michalis Famelis 10.1145/3729383 (Proceedings of the ACM on Software Engineering (FSE))	Jan 2025
DTInsight: A Tool for Explicit, Interactive, and Continuous Digital Twin Reporting Kérian Fiter, Louis Malassigné-Onfroy, Bentley Oakes 10.1109/MODELS-C68889.2025.00030 (MODELS Companion (MODELS-C))	Jan 2025

- Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State of the Practice** Apr 2024
Bentley Oakes, Michalis Famelis, Houari Sahraoui
[10.1145/3638243](#) (ACM Transactions on Software Engineering and Methodology (TOSEM))
- Toward Intelligent Generation of Tailored Graphical Concrete Syntax** Jan 2024
 Meriem Ben Chaaben, Oussama Ben Sghaier, Mouna Dhaouadi, Nafisa Elrasheed, Ikram Darif, Imen Jaoua, **Bentley Oakes**, Eugene Syriani, Mohammad Hamdaqa
[10.1145/3640310.3674085](#) (MODELS 2024)
- Towards Ontological Service-Driven Engineering of Digital Twins** Jan 2024
Bentley Oakes, Claudio Gomes, Eduard Kamburjan, Giuseppe Abbiati, Elif Ecem Bas, Sebastian Engelsgaard
[10.1145/3652620.3688261](#) (MODELS Companion (EDTconf) — Best Short Paper Award)
- Toward a Systematic Reporting Framework for Digital Twins: A Cooperative Robotics Case Study** Jan 2024
 Santiago Gil, **Bentley Oakes**, Cláudio Gomes, Mirgita Frasheri, Peter G. Larsen
[10.1177/00375497241261406](#) (SIMULATION)
- openCAESAR: Balancing Agility and Rigor in Model-Based Systems Engineering** Jan 2023
 Maged Elaasar, Nicolas Rouquette, David Wagner, **Bentley Oakes**, Abdelwahab Hamoulhadj, Mohammad Hamdaqa
[10.1109/MODELS-C59198.2023.00051](#) (MODELS Companion (MODELS-C))
- Machine Learning-Based Fault Injection for Hazard Analysis and Risk Assessment** Jan 2021
Bentley Oakes, Mehrdad Moradi, Simon Van Mierlo, Hans Vangheluwe, Joachim Denil
[10.1007/978-3-030-83903-1_12](#) (SAFECOMP 2021 (Computer Safety, Reliability, and Security))

Publications

- See bentleyoakes.com/publications for a full list of dozens of peer-reviewed publications, including journal articles, book chapters, conference papers, workshop papers, and technical reports/theses.